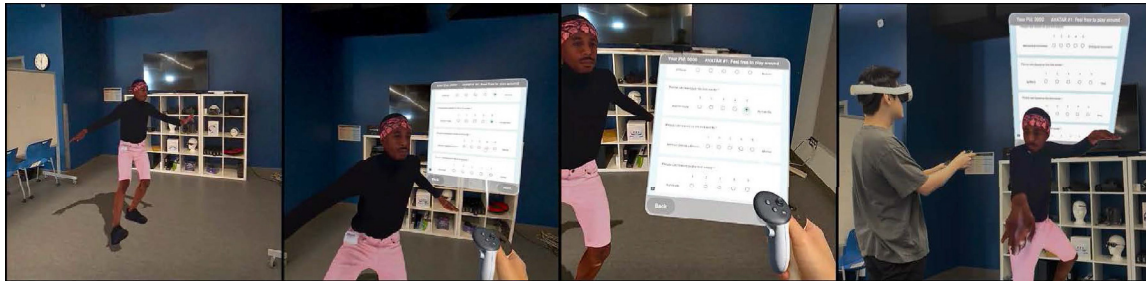


StudyXR: A Framework to Streamline User Study in Immersive Environments

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Figure 1: *StudyXR* teaser: streamlined user study in mixed-reality

ABSTRACT

Immersive and spatial computing technologies such as virtual reality, augmented reality, and mixed reality are bringing about interdisciplinary transformations in multiple fields. As immersive technology continues to rise, the increasing trend of conducting user studies using such technology is obvious. In this paper, we propose several extensions to traditional frameworks, which form our novel framework, *StudyXR*. In addition to technical extensions, an exploratory user study is also conducted based on the user experience and usability of our framework. Both quantitative and qualitative data are collected. Results demonstrate satisfactory effectiveness, making it robust and capable of facilitating user studies in immersive environments.

Index Terms: VR, AR, MR, Games, User Studies, Automation

1 INTRODUCTION

In recent years, immersive and spatial computing technologies such as virtual reality (VR), augmented reality (AR), and mixed reality (MR) are experiencing rapid advancements. The recent introduction of immersive platforms such as Meta Quest 3¹ and Apple Vision Pro² has marked a significant milestone that has elevated the immersive experience to unprecedented heights. Such technologies are bringing about interdisciplinary transformations in multiple fields, including education, healthcare, and e-commerce. The study by Al-Ansi et al. revealed that the practical implications of immersive technologies have the potential to revolutionize the field of education by offering more immersive, interactive and accessible learning experiences [1]. An extensive application of immersive technology has also been found, but not limited to, K12 and higher education, as well as special education, engineering, and medical training [16]. Pons et al. have found a wide range of scenarios in which immersive technologies, such as head-mounted displays

(HMD), played an important role in mental health [10]. Immersive technology has also created a novel way to provide more interactive and personalized experiences to consumers in online shopping [14]. High-quality representations and reconstructions of human beings are used to foster presence and realism in the immersive environment [8]. Visualization and presence have been widely researched and shown to be significant in perceived realism [15]. Miao et al. have proposed that in different contexts, avatar type could tremendously impact user experience [7].

As immersive technology continues to rise, the increasing trend of conducting user studies using such technology is obvious. Within this interdisciplinary field, conducting studies on user perception and experience presents unique needs. Based on a previous work, *StudyVR*, as a comprehensive framework, has addressed several challenges, including the need for specialized equipment, complex coding and data management [6]. However, several limitations of *StudyVR* were also identified. Here we propose an extension to *StudyVR*, which makes our framework *StudyXR*. As an extended version of *StudyVR*, *StudyXR* expands the capability, but also maintains the proven experimental design. A user study is also conducted based on the user experience and usability of *StudyXR*, making it more robust and capable of facilitating user studies in any immersive environment.

2 BACKGROUND AND RELATED WORKS

2.1 Immersive Technologies

In the current literature, VR is defined to be one of the branches of extended reality (XR), which also covers AR and MR. To distinguish these immersive technologies, VR is considered to replace the real world with a virtual one, whereas AR enhances the real world by overlaying virtual graphics onto it; however, MR combines everything, providing a platform where virtual objects and the real world can interact with each other. To achieve these results, head-mounted displays (HMDs) are currently the most prevalent platform. Using stereoscopic lenses, HMDs can create an illusion of three-dimensional depth by simulating and displaying images from different angles that match the distinct views of each eye. While HMDs are often considered the symbol of immersive technology, the core of such technology is, however, the cognitive experience they offer. As display technology has rapidly evolved, in recent years, the emphasis of immersive technology has gradually transi-

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¹<https://www.meta.com/quest/quest-3/>²<https://www.apple.com/apple-vision-pro/>

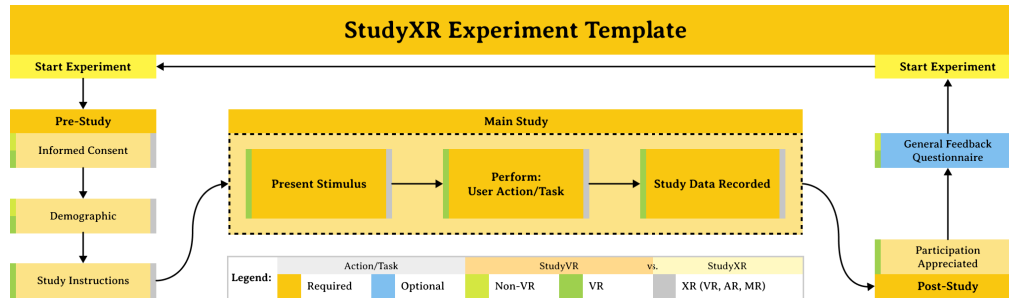


Figure 2: Comparison of experimental framework between traditional *StudyVR* [6] and *StudyXR*

tioned from display advancements to cognitive and perceptual experiences [3].

2.2 User Experience

With a growing emphasis on user experience, a significant amount of research has been documented in the literature. Salagean et al. discussed the importance of the presence of the stimulus [12]. Embodiment and self-identification are suggested to be a crucial key while investigating user experience. Study has also found that the presence of the stimulus can comfort the user, leading to more disclosure of information, which is very important for user studies [9]. Dwivedi et al. also proposed the use of MR, in terms of avoiding presence interruptions, that MR provides a realistic augmentation and interaction between the real world and virtual objects, mitigating the breaks in presence due to awareness of the virtual environment or perception gap [2]. The literature found that in-VR questionnaire is less invasive and more capable of providing reliable self-reporting [11]. Researchers have also argued that in-VR questionnaires provide a more continuous user experience and avoid the disadvantages of post-test questionnaires, especially in studies that require an immediate response after each stimulus [13]. Meanwhile, previous research suggested that familiarity with technology and interfaces can significantly influence user experience [4]. It is worth having common user interfaces other than the stimulus to further foster the outcome of user perception on avatars.

2.3 Related Work: *StudyVR*

As the basis for this study, *StudyVR* provides a novel framework to streamline the conduct of VR user studies. However, several limitations were found to be significant and need attention. In such a study framework, a critical interaction paradigm is that users are required to perform frequent and subtle physical movements. These micro-movements enable participants to explore stimulus from various perspectives, which satisfies the user's curiosity or the expectation of research. In *StudyVR*, although physical movement is freely allowed, due to the distinction between virtual and real environments, virtual locomotion and rotation are still the common way of moving, which is always the main factor in causing cybersickness. Meanwhile, the sense of insecurity due to the potential risk of collision also significantly distracts the user from perceiving the stimulus. As a VR framework, it could potentially create a perceptual gap if the virtual environment is not properly controlled. In the meantime, from a technical perspective, although *StudyVR* aims to provide an easy way to deploy the study setup, it still needs a great amount of effort to build up the server-client connection due to the redundancy in study management between server and client. This configuration in *StudyVR* would also compromise the customizability of this framework, posing an unnecessary architectural complexity. And finally, as a proof-of-concept framework, *StudyVR* still lacks validation of user experience and usability, as these aspects are proved to be crucial to a study framework.

To address these challenges and limitations, we propose our novel framework, *StudyXR*. As an extension to *StudyVR*, it shares the same goals with *StudyVR* as the following:

1. **Easy Study Management:** *StudyXR* aims to provide a simple way for researchers to manage studies, implement approved methodologies, and use any type of stimuli.
2. **Easy Interface:** *StudyXR* also simplifies the interface that allows study designers to easily present their study ideas.
3. **Easy Data Management:** *StudyXR* provides a versatile framework that can integrate with various data collection platforms to fit different data management needs.
4. **Easy XR Deployment:** Using the novel XR technology, *StudyXR* enables easy deployment on any of the XR platforms, ensuring seamless switching between VR, AR, and MR without much effort.

In this study, the addition of an MR system prevents break in presence and the potential risk of cybersickness. It raises user's familiarity by providing more natural interaction. A simplification also leverages customizability and ease of use for researchers. In addition to technical extensions, an exploratory user study is conducted based on the user experience and usability of *StudyXR* to further prove the effectiveness of the extensions.

3 METHODOLOGY

3.1 Technical Extension

As shown in Figure 2, a new MR system has been added to the traditional framework, allowing all parts of the study to be completed in an XR environment. To achieve this, the Meta MR Utility Kit was used to build the connection between virtual objects and the real world. Spatial data were synchronized in real time, creating a global EffectMesh, to ensure proper mapping on virtual passthrough environment. Real-time lighting rendering techniques were used to generate realistic illumination and accurately project dynamic shadows onto the real world. With the passthrough capability, users can see the real world seamlessly just like what they can see without the headset, and this significantly mitigated the sense of insecurity due to the risk of collision. Without the need for virtual locomotion and recentering, the user interface was simplified, leading to an intuitive learning trajectory and less confusion.

Meanwhile, a more concise configuration panel was integrated into Unity's inspector, modularizing study presentation, and replacing the previous *Airtable*³ implementation. This refinement eliminated unnecessary server-client setup while maintaining a streamlined study management approach, providing more space for developers to customize their study.

³<https://www.airtable.com>

To validate extensions to the traditional framework, a user study focused on the user experience and usability of *StudyXR* was carried out.

3.2 Apparatus

Due to *StudyXR* being a highly streamlined framework, the experimental process required minimal equipments, reducing technological barriers while keeping the space for further extension if needed. In this user study, a *Meta Quest 3* served as the primary tool to present the full procedure to the user, while the researchers used an *iPad Pro* to monitor and capture the scene from the user's viewpoint. A 5-meter by 5-meter area was also designated for the experiment.

3.3 Participants

Our exploratory user study recruited eight participants by posting recruitment invitations in the game development and UI/UX design communities. Five males and three females are reported between 18 and 34 years old, with the majority ($n = 5$) in the 25-24 age range. The sample demonstrated a wide-ranging gaming experience, with participants showing varied levels: 2 identified as intermediate players, 2 as experts, 2 as advanced, 1 as casual, and 1 as a beginner. Participants used primarily PC as their main gaming platform ($n = 7$), while additional platforms such as mobile devices ($n = 4$) and handheld devices ($n = 3$) were also used. In particular, 62.5% of the participants ($n = 5$) reported having short-sightedness or farsightedness, and only one participant reported being tested for visual impairment.

3.4 Procedure

The experimental procedure was structured as depicted in Figure 2, aiming to mimic a practical scenario since the primary emphasis was on post-use experience and feedback. Upon completion of bringing the participants to the experiment area, the researchers first set up the scene for the participants, including adjustments to the overall lighting direction, the position of the stimuli to be presented, and the unique ID for each user. The participants were then introduced with the settings of the experiment, the devices to be used, and informed consent. Once the connection between Meta Quest 3 and iPad Pro was successfully built, the participants were asked to put on the headset and start the study.

The participants would first encounter the one-pager tutorial with only three steps. After they got familiar with the controls, the webview renderer displayed a demographic questionnaire, allowing the user to complete the questionnaire by simply clicking and drag. Two virtual avatars were then presented by Volumetric Video Streaming (VVS) in fixed order as stimuli [5]. The participants were free to take their time to play with the virtual avatars and could continue whenever they felt ready. Two short surveys of partial Godspeed Questionnaire, regarding the two virtual avatars respectively, were displayed after the perception session. Participants were asked to complete the surveys as quickly as possible to prevent the fatigue effect since perception ratings were not the focus of this study. Upon completing the mock sessions, participants were required to fill out an AttrakDiff Questionnaire along with a Simulator Sickness Questionnaire (SSQ) for quantitative feedback. Once all questionnaires were completed, participants removed the headset and participated in a semi-structured interview to provide qualitative feedback. They were systematically asked key aspects of their perception on the shadows of the avatar, *StudyXR*'s capacity to create a sense of realism, and the overall usability and experience of *StudyXR*. This session ended with open questions, which allowed participants to discuss any topic they wished. After encountering the whole study, participants were thanked for their participation. No compensation was provided to participants in this study, pre-

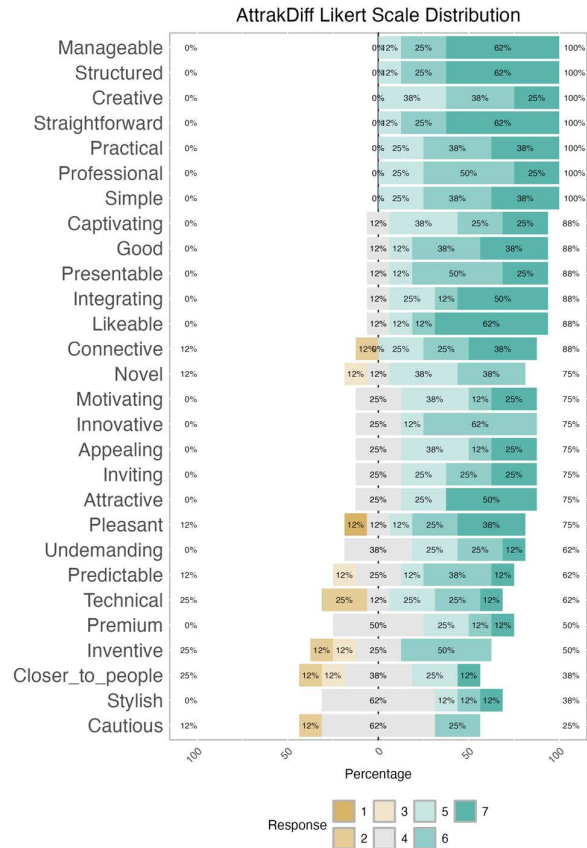


Figure 3: Results of AttrakDiff, showing each item of the questionnaire, ordered descendingly from top to bottom.

venting potential financial incentives that could influence research results.

3.5 Design

As an exploratory user study using single-system evaluation, since the perception ratings of virtual avatars in this study were not interested, no counterbalance or randomization was utilized.

4 RESULTS AND DISCUSSION

4.1 AttrakDiff

The AttrakDiff Questionnaire was used to validate the usability, user experience, and appearance ratings of *StudyXR* with four dimensions, which are **Pragmatic Quality (PQ)**, **Hedonic Quality - Identity (HQ - I)**, **Hedonic Quality - Stimulation (HQ - S)** and **Attractiveness (ATT)**. Each dimension has seven items, forming a list of 28 questions. The order of the questions was randomized and reverse coding was performed. Figure 3 shows the overall distribution of responses to the AttrakDiff Questionnaire. The mean and standard deviation were calculated for each item and each dimension as shown in Table 1 and Figure 4.

4.2 Simulator Sickness Questionnaire

SSQ was used to assess user's physical discomfort with 16 items. Each item had a rating of 1 to 4, indicating from none to severe. The response distribution is shown in Figure 5, with only a few reported discomfort on vision and fatigue (Overall $M = 1.23, SD = \pm .49$).

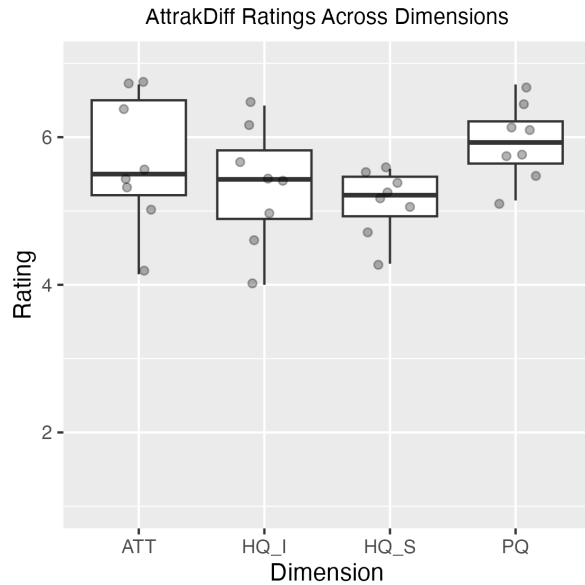


Figure 4: Results of each dimension of AttrakDiff

AttrakDiff Dimension	Mean	Standard Deviation
PQ	5.93	±.52
HQ - I	5.34	±.80
HQ - S	5.13	±.45
ATT	5.66	±.91

Table 1: Mean and SD for each AttrakDiff Dimension

4.3 Qualitative Comments

The study was concluded with a semi-structured interview. The qualitative comments provided insights on different aspects. Participants frequently commented on the realism of avatars ($n = 6$); although this was not the focus of this study, these comments nevertheless provided empirical validation of the effectiveness of the framework. Interestingly, many participants reported that they “did not consciously notice” the realistic shadows by real-time lighting ($n = 4$). In terms of usability of the framework and design of the interface, all participants appreciated the ability to move and scale the floating panel for navigation and the questionnaire ($n = 8$). Participants also highlighted the framework as “easy to learn” with a minimal interface design. Some participants expressed concerns about the responsiveness of the buttons on the panel, suggesting the need for clearer feedback mechanisms, such as sound or haptic sensations ($n = 3$). A few participants ($n = 2$) mentioned that the prompt text in the upper right corner of the panel, intended to guide users, was often “overlooked”. Suggestions were made for changing the color or size of the text to increase visibility. Concerns were raised about the limitations of using Google Forms for questionnaires, suggesting the need for alternative platforms ($n = 5$). For physical discomfort, some participants ($n = 3$) mentioned the weight of headset being too heavy that could potentially lead to fatigue or eye strain.

5 DISCUSSION AND CONCLUSION

In this study, several extensions were proposed and developed to the traditional *StudyVR* framework, forming the new *StudyXR* framework, including the addition of MR systems, new mechanism for locomotion, real-time lighting, EffectMesh and minimal interface

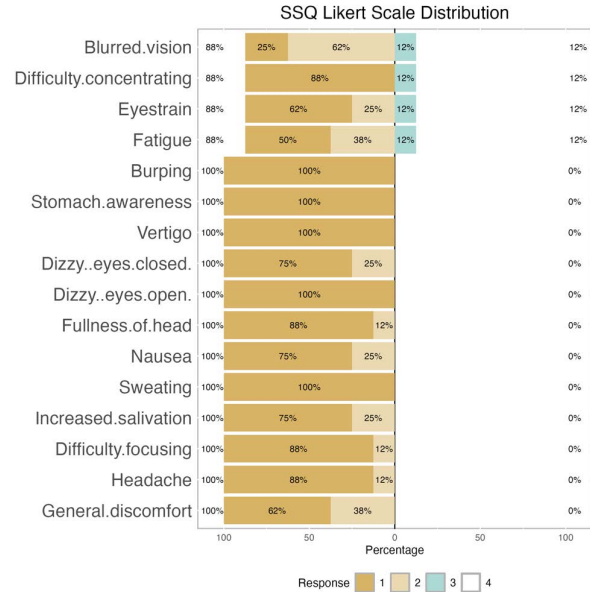


Figure 5: Responses of SSQ, showing each item of the questionnaire, ordered descendingly from top to bottom.

design. The AttrakDiff results reveal strong positive ratings across all dimensions, especially in PQ and ATT. These ratings indicate that *StudyXR* not only meets functional needs with decent usability but also provides an engaging and appealing user experience. Only a small portion of the discomfort reported from the SSQ responses, mainly limited to vision and fatigue issues, was also explained by the qualitative comments, namely the weight of the headset. In general, the effectiveness of the framework in mitigating cybersickness is shown by decent SSQ results. Qualitative feedback from participants not only provides additional validation of *StudyXR* but also reveals areas for potential enhancement. The description of *StudyXR* as “easy to learn” aligns with the original goal of providing an accessible research tool for both researchers and participants. An interesting finding of the “unconscious” nature of the real-time lighting effects also validates successful integration of realistic rendering, which prevents drawing attention to the technical implementation. While *StudyXR* demonstrates satisfactory effectiveness, usability, and user experience, several limitations should be acknowledged. First, the sample size ($n = 8$) is relatively small, although appropriate for an exploratory study. Future research should have a larger and more diverse sample to further validate the findings. From a technical perspective, some potential enhancements include additional feedback mechanisms, a higher visual hierarchy in the interface design, and a more XR-friendly questionnaire platform.

In conclusion, *StudyXR* successfully extends the capability of the traditional *StudyVR* framework. The extensions simplify the overall architecture of the framework. In addition, *StudyXR* demonstrates adequate usability and user experience and minimizes cognitive load and physical discomfort. Together with all these extensions, *StudyXR* maintains the proved experimental design of the traditional framework, making it a robust and user-friendly tool for researchers to streamline user studies in immersive environments. With these improvements, *StudyXR* is positioned as a valuable contribution to the field of XR research. As XR technologies continue to evolve, frameworks like *StudyXR* will become more crucial in supporting robust research, ensuring that both real-world applicability and user ease are preserved.

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